

How did the COVID-19 pandemic affect CO2 emissions in major European cities, and what long-term impacts might this have on future emissions trends?

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I. INTRODUCTION

The COVID-19 pandemic, an unprecedented global health crisis, has had far-reaching impacts on human life and the environment. One of the most significant and unexpected environmental changes observed during the pandemic was a noticeable drop in greenhouse gas emissions, particularly CO₂, due to reduced human activity. Major European cities, characterized by their high population density and industrial activity, faced varying degrees of lockdown measures that significantly influenced energy consumption, transportation, and overall emissions patterns.

The pandemic has presented a unique opportunity to study the intricate relationship between human activity and environmental impact. By analyzing changes in CO₂ emissions during different phases of the pandemic, such as lockdowns and reopening periods, we can gain insight into the factors that contribute to emissions reductions and increases. This analysis is crucial to understanding not only the immediate effects of the pandemic but also the potential for sustainable recovery and long-term environmental policy making.

The study of CO₂ emissions in major European cities is particularly relevant due to the region's significant contribution to global emissions and its role as a leader in environmental sustainability initiatives. The findings of this research can inform urban planning and policy decisions aimed at achieving lower emissions, after the pandemic, aligning with broader climate action strategies.

This literature review combines current knowledge on the topic, highlighting key findings from recent studies and identifying gaps in the research. It examines the methodologies used to analyze the CO₂ emissions data, the factors influencing emissions changes during the pandemic, and the potential long-term impacts on emissions trends. By reviewing the existing literature,

this paper aims to contribute to understanding the lasting impact of the pandemic on the environment and its implications for future sustainability efforts. This paper aims to explore the impact of the COVID-19 pandemic on CO₂ emissions in these cities and to discuss the potential long-term implications for future emission trends.

II. RELATED RESEARCH

The COVID-19 pandemic has led to numerous studies investigating its environmental repercussions, particularly with respect to greenhouse gas emissions [3]. The drastic changes in human behaviour due to lockdowns and travel restrictions provided a unique opportunity to observe the impact on CO₂ emissions. However, the focus on major European cities has been less extensive, warranting a more nuanced analysis of urban-specific data [7]. Although these studies have provided valuable information on the immediate effects of regulating carbon emissions, they have often fallen short in terms of long-term predictive modeling. Furthermore, the application of machine learning techniques to forecast future CO₂ emissions has been underexplored [2], [6].

A. Global and National Studies

Global studies have predominantly used satellite observations, government reports, and real-time data to estimate emission reductions during the pandemic [4], [5]. These studies have reported a temporary decrease in daily global CO₂ emissions, primarily attributed to the decrease in surface transport and industrial activities [4], [5]. However, while these findings are valuable, they often do not account for differences in emission responses in various urban areas [8]. This gap highlights the need for a more detailed analysis that takes into account the unique characteristics and responses of urban areas to the pandemic.

B. European City-Specific Studies

A handful of studies have centered on European cities, using direct observations and bottom-up approaches to provide a more detailed analysis [7], [8]. These studies have revealed significant variations in emission reductions among different urban districts, which can be attributed to the stringency of lockdown measures and the underlying urban structure [8]. Despite these insights, there is still a lack of research on forecasting long-term emission trends and developing tailored policy recommendations for European cities [10].

C. Methodological Approaches

Researchers have drawn on a variety of methods to study the environmental fallout from the COVID-19 pandemic, from classic statistical approaches like ARIMA to cutting-edge machine learning models such as LSTM [1], [2]. These techniques have proven critical for understanding the short-term effects of the pandemic on our environment. However, there's a noticeable gap when it comes to using these tools for long-term predictions and shaping environmental policy [6]. This gap allows us to explore the need for further research on how these methods can help predict future emission patterns and develop effective environmental strategies.

D. Summary of Current Literature

Current research on the impact of the COVID-19 pandemic on CO₂ emissions has primarily utilized statistical analysis and direct observations to analyze immediate changes during the pandemic [3], [4]. However, these studies have generally not employed machine learning approaches, which could offer more sophisticated predictions of future emissions trends [2], [5]. Additionally, while some studies have presented broad policy recommendations, they often lack specificity and are not tailored to the unique needs of major European cities [3], [4]. A significant gap in the literature is the lack of long-term predictive modeling that extends beyond the immediate effects of the pandemic [6]. This is particularly important given the urgent need for strategies that are attuned to the unique characteristics of European urban environments [7], [8].

E. Datasets & Results

For this literature review, a variety of datasets were utilized to assess the impact of the COVID-19 pandemic on CO₂ emissions in major European cities. The European Environment Agency's Greenhouse Gas Emissions Database provided a complete overview of emissions across Europe. National Inventory Reports (NIRs) offered insights into country-specific emissions, while Google Mobility Reports delivered real-time data on mobility patterns during the pandemic. Additionally, the Global Carbon Project Data and the Oxford COVID19 Government Response Tracker

(OxCGRT) were fundamental in understanding global emissions trends and the stringency of lockdown measures, respectively. Urban Eddy-covariance datasets have been used for detecting short & long-term changes in carbon fluxes due to their ability to observe gas momentum between the atmosphere & ecosystems. Providing a better comprehension of carbon cycles in cities. The findings from various research papers highlighted significant changes in CO₂ levels during the pandemic.

Notably, global greenhouse gas emissions experienced a notable drop, particularly in April 2020, with a peak reduction of 17% [3]. Studies also indicated a temporary reduction in daily global CO₂ emissions during the pandemic largely due to changes in surface transport. The estimated decrease in 2020 annual emissions ranged from -4.4% to -8.0% [4]. Furthermore, direct observations revealed substantial reductions in CO₂ emissions in European urban districts during lockdowns, with reductions ranging from 5% to 87% [8]. Predictive modeling using SARIMA demonstrated the potential for forecasting future emissions trends, with the post-COVID-19 model achieving the highest accuracy with a MAPE of 0.09 [6]. These findings accentuate the potential of machine learning techniques in predicting CO₂ emissions and suggest that future research should focus on developing and refining these models for better environmental policy-making.

F. Our Distinct Approach and Contribution

Our research addresses these gaps by focusing on the application of machine learning algorithms to model and predict CO₂ emissions in major European cities [6]. By leveraging machine learning, we aim to conduct a more detailed analysis of the pandemic's impact and enhance the accuracy of future emissions forecasts. This approach is expected to yield more precise predictions, providing valuable insights for urban planning and policy-making [2]. Our work will contribute to the field by employing machine learning models such as K-Nearest Neighbors (KNN) and Long Short-Term Memory (LSTM) networks to analyze CO₂ emissions data [6]. These models will help identify complex patterns in the data that traditional statistical methods may overlook. Furthermore, by concentrating on European cities, our research will offer targeted policy recommendations to support these urban centers in achieving more sustainable futures [7].

III. METHODOLOGY

This section details the methodological approach of this study, outlining the procedures employed to analyze the impact of the COVID-19 pandemic on CO₂ emissions in major European cities and discuss potential long-term implications for future emissions trends.

A. Research Design

Our study employs a mixed-methods design, integrating quantitative analysis of CO₂ emissions data with qualitative assessment of policy implications. This approach allows us to use the strengths of both quantitative and qualitative methods, providing a comprehensive understanding of the impact of the COVID-19 pandemic on CO₂ emissions in major European cities.

B. Data Sources

Data will be sourced from the European Environment Agency's greenhouse gas emissions database, national inventory reports, and satellite observations.

C. Data Collection

Emissions data will be collected for the period preceding, during, and following the pandemic lockdown periods. This longitudinal data collection will allow us to analyze changes in emissions over time, capturing the immediate effects of lockdowns and the recovery phases.

D. Participants/Sampling

The study focuses on a purposive sample of ten major European cities, selected based on population size and industrial activity.

E. Variables

The independent variable is the stringency of lockdown measures, while the dependent variable is the change in CO₂ emissions. Control variables include economic indicators and urban characteristics.

F. Instruments and Tools

CO₂ flux data will be analyzed using Python with libraries such as Pandas and Scikit-learn. The policy documents will be qualitatively evaluated using content analysis.

G. Data Processing and Analysis

Data will be cleaned and preprocessed to handle missing values and outliers. Emissions trends will be analyzed using time-series analysis and machine learning models.

H. Machine Learning Models

We will employ ARIMA for time-series forecasting and LSTM networks to predict future emission trends. To provide context for our choice of models, we present a summary of machine learning models used in related studies in **Table I**. This comparison highlights the diversity of approaches in the field and underscores the rationale behind our selection of models.

I. Procedure

The study will follow a chronological procedure from data collection in 2020 to analysis and reporting in 2023.

J. Ethical Considerations

The study adheres to ethical standards for environmental data use and ensures confidentiality of policy-related information.

K. Limitations

The methodology acknowledges limitations due to data availability and the dynamic nature of urban emissions.

IV. RESULTS & ANALYSIS

This section presents the results of the analysis of the CO₂ emissions data. It includes graphical representations of the data and a detailed discussion of the findings.

Among the articles, the consensus of their results is that while the COVID-19 pandemic greatly affected CO₂ emissions, the effects highlighted during this period are temporary & not sustainable. As a result of the lockdown, many daily activities were abruptly stopped to contain the spread of the virus. This measure was accompanied by a reduction in European carbon emissions, which researchers have identified several of the main contributors to CO₂ emissions.

A. Emissions in Europe

Due to geographical differences between countries in Europe, carbon emissions are all affected by different factors. These could include When comparing the results of different municipalities, it is crucial to remember that they will vary in size across different countries. Scandinavian countries tend to generate more emissions due to larger administrative units.

A Lorenz curve provided in this paper showcases the "contribution of municipalities to the total European emissions". We can see from it that the top 10 municipal cities produce 33.4% of total European emissions. However, this does not mean that these cities are densely populated. These high carbon emissions can be the result of industrial work. Examples of this can be seen in Gotland where there is one large cement facility [7].

Utilising the Tobit panel model, we can see that good governance plays a major role in reducing carbon emissions. The governance elasticity is -0.23, which is the largest elasticity for CO₂ emissions displayed in the study. As CO₂ drops with each increase in governance both short-term & mid to long-term policies must be implemented to help improve the carbon emissions in municipal cities while maintaining the effects seen during the COVID-19 pandemic [10] [11].

When comparing changes in the spatial distribution of annual CO₂ emissions from 2019 to 2020 & 2016 to 2019, Europe was part of the group that had higher carbon emissions in 2019 compared to 2016, while in 2020 it had higher carbon uptakes than in 2019 [9].

TABLE I
SUMMARY OF MACHINE LEARNING MODELS USED IN RELATED STUDIES

Paper	XGBoost	Linear Regression	K-means	GBDT
6	X			
2		X	X	
1			X	

B. Traffic in Europe

When reading the implications of carbon emissions in Europe, the largest contributors to carbon emissions in 2019 were commutes executed by students & staff. This is supported by the results acquired during the comparison with city-scaled activity data. During the lockdown period, CO₂ emissions in residential areas did not have any notable increases, despite a 20% rise in time spent at home [11].

A link was observed between reduced road traffic emissions & lower measured CO₂ emissions in all European cities. Throughout the transition between lockdown to post-lockdown, the correlation decreased until its disappearance in the second post-lockdown where it had reached pre-pandemic measurements. The results indicate that emissions caused by traffic are one of the main contributors to CO₂ emissions in Europe compared to emissions generated in a households. [8]

C. Air Pollution in Europe

As discussed in the previous segment, traffic along with activities similar to industrial production was halted. Due to this air pollution caused by these activities decreasing, during February 2020 in Lyons, France, air pollution was recorded at 36.8 μ g/m³. Between March 17th to May 11th, 2020 it dropped to 12.0 μ g/m³.

Throughout this period a decrease in NO₂, NO & CO by 67%, 23% & 53% whereas an increase in O₃, PM₁₀, PM_{2.5} by 105%, 23% & 53% was discovered. While there is a positive decrease in harmful chemicals in the air, there has also been an increase in chemicals that are harmful to the human body. Unfortunately, it is not clarified as to what distance O₃ is located at, since it does not hurt humans when located at a high level [11].

When compared to a study done in Barcelona city, shows that PM_{2.5}, PM₁₀, NO₂, CO & SO₂ reduced were by 19%, 32%, 54%, 20%, & 25%. This could indicate that other factors in Lyons increased PM₁₀ & PM_{2.5}.

D. Potential Strategies for Europe

Although the COVID-19 pandemic showed a positive influence in Europe in numerous studies, the decline in these effects afterward has also been highlighted. In addition to this, there have been negative reports about its effects on soil pollution due to protective gear,

and household & plastic waste. As a result, the effects gained during this period must be preserved.

Governments should implement short-term strategies while developing mid to long-term strategies. These strategies in development should work towards meeting renewable energy targets. Implementing adjustments to regulations such as decreasing access highly polluting cars have to clean air locations or implementing zero carbon emission targets for new buildings can allow for imperishable effects on carbon reduction [11].

V. CONCLUSION

This literature review has explored the impact of the COVID-19 pandemic on CO₂ emissions in major European cities, highlighting both immediate effects and potential long-term trends. Our study addresses the gaps in current research by applying machine learning algorithms to predict future emissions trends, focusing on a sample of major European cities.

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